Review of "Memory Rank and Liftability of Transition Families Without a Generator: A Projection Theory for the Indivisibility Deficit"

Overall Assessment

This is one of the more coherent and mathematically disciplined papers in the series. It has a clear central construction: given a finite family of two-time stochastic kernels that fails Chapman-Kolmogorov composition, the paper measures how closely that family can be represented by the observable compression of a higher-dimensional Markov process. The resulting profile:

$$h_d(V) = \mathrm{dist}(V, \mathcal{F}_d)$$

is a natural and potentially useful way of quantifying the hidden-state dimension required to explain an apparently non-Markovian transition family.

The paper also has a sensible overall progression. It first identifies the Chapman-Kolmogorov defect and derives its four-time coherence identity; it then shows why minimization over invertible re-descriptions cannot yield a meaningful invariant; it replaces that failed approach with distance to a fixed hidden-state model class; and it develops compactness, semialgebraicity, monotonicity, stability, realization results, category dependence, and numerical examples. The distinction between exact realization and membership in the closure of the model class is now stated unusually carefully, and the limitations of the numerical optimization are also acknowledged more honestly than in several of the earlier manuscripts.

My overall recommendation would nevertheless be **major revision**. The principal results appear to contain a substantial mathematical core, but the paper currently presents several interpretations and contributions more strongly than the established results warrant.

The central liftability profile is clearly defined and mathematically natural. The closed model sets give compactness and attainment, their nesting makes the profile monotone in hidden dimension, and the 1-Lipschitz stability result provides a useful link to statistical estimation. The finite-dimensional history-state realization, the marginal-consistency obstruction, the careful distinction between computed upper bounds and proved lower bounds, and the discussion of exact versus closed-model liftability are further strengths.

Overall Conclusion

The paper has a viable core and is substantially stronger than a purely speculative proposal. The metric-projection formulation is clean, the realization theorem is useful, and the uniform marginal-consistency obstruction is probably the most concrete original component. The main revisions required are conceptual calibration and positioning: the manuscript should separate genuinely new model-specific results from general facts about compact semialgebraic model classes, formalize or reduce the categorical and cohomological terminology, clarify what the proposed rank measures, and make the numerical evidence reproducible.

Major Comments

The meaning of "without a generator" should be qualified throughout. Failure of the Chapman-Kolmogorov identity excludes a time-local Markov evolution or time-ordered-exponential generator on the

same observed state space. It does not exclude hidden-state generators, memory-kernel equations, or other nonlocal dynamical representations. Phrases such as "no generator reproduces them" are therefore broader than the mathematical statement established here. The title, abstract, significance statement, and introduction should consistently specify "no time-local Markov generator on the observed state space," or an equivalent formulation.

The connection with quantum non-Markovianity is too compressed. The manuscript identifies the classical Chapman-Kolmogorov defect with the object used to measure failure of divisibility in quantum-information theory. In the quantum setting, however, CP-divisibility concerns the existence and complete positivity of intermediate propagators and is not simply the failure of an arbitrary two-time conditional family to satisfy the classical identity in equation (2). The analogy may be useful, but it should be formulated as an analogy rather than an identification, with the precise correspondence and its limitations stated.

The abstract overstates what the numerical results establish.

The statement that the profile "attains zero exactly at the generating dimension" suggests that minimality has been proved. The computations establish exact vanishing at the known generating dimension and provide numerical evidence that the profile remains positive below it. Since the lower-dimensional optimizer values are only upper bounds, the abstract should say this explicitly, for example:

> "the profile is shown to vanish at the generating dimension, with numerical evidence that this dimension is minimal in the examples considered."

The use of "dilation category" requires either formalization or replacement. The manuscript currently defines classes of lifts but not a category: no objects, morphisms, identities, or composition law are specified. This becomes important because the abstract, introduction, Definition 4.1, and later open problems attach substantive meaning to category dependence and categorical morphisms. If no formal category is needed for the results, "dilation class" or "model class" would be more accurate and would avoid creating an unnecessary foundational obligation.

The cochain and complex terminology exceeds the formal construction currently provided.

Proposition 2.2 gives a correct and useful four-time coherence identity. However, describing V as a 1-cochain, Δ_V as a 2-cochain, and δ_V as a twisted coboundary suggests an actual cochain complex. The manuscript does not define the complete graded cochain spaces or show that an appropriate coboundary operator squares to zero; indeed, δ_V is twisted by a generally non-flat family. The paper also postpones the higher identities and associahedral indexing.

The safest formulation would be to call equation (5) a **Bianchi-type coherence identity** and present the cohomological interpretation as a possible future formalization. The title "deficit complex" in Section 2 should likewise be reconsidered unless the complex is fully defined.

The gauge-trivialization theorem depends on a very broad and potentially nonphysical notion of re-dressing.

The algebra of Theorem 3.3 appears sound, but the interpretation needs qualification. The matrices g_i preserve total mass, while their inverses need not preserve positivity or map the probability simplex into itself. Admissibility only requires the particular transformed kernels V^g to remain stochastic. Thus these transformations are not necessarily stochastic isomorphisms or genuine changes of probabilistic coordinates.

This matters because the manuscript concludes that "the gauge route to an invariant is empty." What is proved is that it is empty under this particular group of mass-preserving invertible linear transformations with family-dependent admissibility. More restrictive and probabilistically natural transformations-such as invertible stochastic maps with stochastic inverses, which essentially reduce to permutations-would not permit the same collapse. The conclusion should therefore be stated relative to the declared re-dressing class rather than as eliminating every natural gauge-based construction.

The geometric invariant theory language in Remark 3.5 should be treated cautiously. The explicit orbit-closure result is clear and sufficient. Calling every point "unstable" in the sense of geometric invariant theory is less clearly justified, particularly because the action is non-reductive, the domain is the constrained stochastic set, and the orbit closure approaches a reset locus rather than the origin of a specified

linear representation. Unless a precise GIT definition and theorem are supplied, this should be described more modestly as an analogy with orbit degeneration.

The metric cannot always be said to be weighted by "the law of the process." The weights π_i are said to be supplied by the law of the observed process. Yet the ambient space includes families that are not induced by any joint process, and later the marginal-consistency obstruction is used precisely to identify such families. For a general V, there may be no process law compatible with all its kernels. The weights should instead be introduced as arbitrary positive reference, sampling, or empirical occupation weights; when a compatible process exists, its marginals may be used as a particular choice.

The claimed novelty of the projection construction needs better positioning. Once $\mathcal{F}_d$ is defined as the closure of a finite-dimensional hidden-state model class, compactness, attainment of distance, stability of the distance function, and semialgebraicity are largely standard consequences. The introduction presently presents the whole package as a new invariant-theoretic construction. It should identify more sharply what is genuinely new here-for example, the particular transition-family model sets, the marginal-consistency lower bound, or the finite-grid realization statement-and distinguish those contributions from general metric and semialgebraic facts.

The term "memory rank" needs early qualification as a closed-model, category-relative quantity. The abstract does mention the declared category and the closed model class, but the significance statement describes the minimal vanishing dimension more directly as "the memory rank." At this point a reader could reasonably interpret it as an intrinsic dimension of the physical process. It is actually the smallest hidden dimension whose **closed compression class** contains the specified finite-grid family. It can vary with the grid, model class, treatment of time, and use of closure. Those qualifications should appear at the first informal introduction of the term.

The opening claim that Chapman-Kolmogorov families are "exactly" flat transports over the complex of time intervals reads partly as a geometric theorem, although flatness is subsequently defined to mean the Chapman-Kolmogorov property. It would be clearer to present this explicitly as terminology or interpretation.

The zero-stratum description is correct but partly definitional: because $\mathcal{F}_d$ is defined as the closure of the compression image, $h_d(V) = 0$ is equivalent to membership in that closed set. The genuinely substantive results are the structure of $\mathcal{F}_d$, its relation to process-induced families, and finite termination for genuine processes. The abstract could emphasize those rather than presenting the zero-stratum equivalence itself as a major theorem.

The description of h_d as "computable" should be qualified. It is defined by a finite-dimensional semialgebraic optimization problem and is therefore computable in principle, but the paper does not provide a generally tractable global algorithm. The later multistart optimization produces feasible numerical upper bounds rather than certified global solutions.

Remark 2.3 speculates about higher coherence identities and homotopy-associative structures but does not use them. This material currently makes the result sound more developed than it is. It could be shortened to a brief observation or moved to the open-problems section.

It may help to say explicitly that the re-dressings form an action on the larger affine space of column-mass-preserving matrix families, while admissibility restricts the portion of an orbit lying inside the stochastic domain. That would make the use of "orbit" more precise.

The notation uses π_i for the fixed metric weights earlier, while the lift begins with an initial distribution π and produces hidden laws $p_i(\theta)$. Although formally distinguishable, the notation is dense and could cause confusion once observed laws μ_i are also introduced. Renaming the fixed metric weights would improve readability.

Proposition 4.4 appears correct, but the proof could be streamlined considerably. Splitting one hidden state into two identical copies is a valid way to embed a d-state lift into dimension $d + 1$, and the intertwining identities preserve the observable compression. However, the proof is notation-heavy relative to the fairly simple idea. It would help to state first that R splits the chosen state, P merges the copies, and $PR = I_d$, before presenting the detailed calculation.

Several properties of $\mathcal{F}_d$ and h_d are standard consequences of their definitions. Compactness follows because $\mathcal{F}_d$ is defined as a closed subset of compact $\mathcal{V}$; attainment follows from compactness; the distance is 1-Lipschitz for any fixed nonempty closed set; and the zero set of the distance is the closed set itself. These are useful facts, but the paper sometimes presents them collectively as if they constituted a substantial new theory. The manuscript should distinguish these general metric facts from model-specific results such as nesting, the marginal-consistency bound, and the realization theorem.

The semialgebraicity result should clarify the role of the finitely many labelings. Proposition 4.5 is plausible: for each fixed labeling, the compression is rational on a semialgebraic domain with positive denominators, and the finite union of its images is semialgebraic. It would nevertheless improve precision to define the parameter space separately for each labeling and then explicitly take the finite union. As written, Θ_d^+ sometimes appears to be a single parameter space even though ϕ is discrete.

Remark 4.7 gives consistency of the plug-in profile, not yet a full estimation theorem. The Lipschitz bound immediately transfers norm consistency and upper convergence rates from $\widehat{V}_N$ to $h_d(\widehat{V}_N)$. That is useful, but calling it an "estimation theorem in embryo" is more accurate than the stronger claims made in the introduction. No sampling model, finite-sample error bound, treatment of estimated weights, or distributional theory is supplied. The manuscript should maintain that restrained description throughout.

The numerical evidence does not establish that the relevant projected points genuinely lie on the boundary of $\mathcal{F}_d$. Remark 4.9 says that the computations indicate genuine boundary contact. However, finding a fitted parameterization with extremely small block mass does not exclude the possibility that the same compressed family has another representation with strictly positive block masses. Nor does it establish that the limiting family lacks an exact positive-mass lift. Boundary contact should therefore be described as evidence within the fitted parameterizations unless an intrinsic boundary test or nonexistence argument is provided.

The distinction between closed-model rank and exact realization is essential, but the terminology remains potentially misleading. Definition 5.1 correctly states that $h_d(V) = 0$ means arbitrary approximation by positive-block-mass lifts, not necessarily an exact d-state lift. Consequently, $\mathrm{rk}_{\mathrm{mem}}$ is a rank of the compactified or closed model class. This is mathematically legitimate, but readers will normally understand "minimal lift dimension" as an exact realization dimension. The notation and surrounding prose should consistently include the qualifier "closed-model memory rank," at least until a compactified lift category realizing all of $\mathcal{F}_d$ is constructed.

The marginal-consistency obstruction is useful, but it tests only a limited necessary condition. Theorem 5.6 is one of the most concrete results in the paper: if no compatible family of one-time marginals exists, then the family remains uniformly separated from every hidden-state model class. The manuscript correctly states that the converse fails. It would help to emphasize that $\kappa_c(V) = 0$ does not establish the existence of a common joint distribution, even before hidden-Markov realizability is considered. Thus the certificate detects pairwise marginal inconsistency rather than non-Markovian memory itself.

Section 5.2 should not be described simply as "sufficiency." Theorem 5.7 assumes that V is already process-induced, meaning that a stochastic process realizing all its two-time conditional kernels exists. Under that assumption, the construction gives a Markov lift by augmenting the state with the complete observed history. This is a valid realization theorem, but it is not a sufficiency criterion stated directly in terms of verifiable properties of V. Renaming the subsection "Realization of process-induced families" would more accurately describe what is proved.

The history-state realization should be positioned as a standard state-augmentation construction. Theorem 5.7(a) is mathematically useful because it gives an explicit finite bound, but its basic mechanism-making the full history into the current hidden state-is standard. The novelty, if any, lies in applying it to this family-of-conditionals framework and deriving the consequences for h_d and the closed model sets. The paper should acknowledge the standard nature of the construction and explain which part of the theorem is new.

The "homogeneous" realization contains an implicit clock.

The proof homogenizes the lift by placing histories of different lengths in disjoint layers. A single transition matrix can then apply the correct rule because each layer is occupied at only one time. This is algebraically

homogeneous, but the hidden state records the time index. It therefore differs from a stationary latent mechanism without time augmentation.

This qualification matters for the interpretation that "time-inhomogeneity can substitute for hidden dimension." Part of that exchange rate arises because the homogeneous category is permitted to reproduce inhomogeneous behaviour by paying for a hidden clock. The manuscript should state this explicitly when interpreting the two ranks.

The universal realization bound is finite but very crude and depends on the grid length.

The bound

$$D \leq \sum_{k=0}^{m} n^{k+1}$$

grows exponentially with the number of observation times because it stores complete histories in distinct time layers. This does establish finite termination for process-induced families on any fixed finite grid, but it does not yet show that the proposed memory rank captures a compact intrinsic memory dimension as the grid is refined. That limitation should be mentioned alongside the theorem rather than only deferred to the open problems.

Definition 5.9 introduces a mathematically valid distinction, but the relationship between its two open questions is overstated. Constructing a compactified lift class whose compression image equals each $\mathcal{F}_d$ would clarify exact realization at fixed d. It would not by itself establish that the increasing union $\bigcup_d \mathcal{F}_d$ is closed or that every asymptotically liftable family has finite rank. These are related boundary questions, but one does not automatically resolve the other.

Section 6 currently gives generic projection geometry rather than model-specific first-order theory. The normal-cone condition, orthogonality to a smooth stratum, almost-everywhere uniqueness of projection, and the ED-degree observation are standard results for distance to closed or semialgebraic sets. The paper does not calculate the tangent or normal cones of $\mathcal{F}_d$, characterize its strata, determine when the compression map is regular, or derive explicit first-order equations for the lift parameters. Therefore the description that first-order theory "is developed" is too strong. Either model-specific content should be added or the section should be presented as the standard geometric framework available for future analysis.

The projection residual is not the Chapman-Kolmogorov deficit.

After equation (9), the manuscript says that "all of the deficit that the stratum's directions could absorb has been absorbed." However,

$$V - W^*$$

is the residual between the observed family and its nearest model point, whereas

$$\Delta_V(t, s, r)$$

is the nonlinear failure of Chapman-Kolmogorov composition. They are different objects living in different spaces. This sentence should refer to the "projection residual" or "model mismatch," not the deficit.

The almost-everywhere uniqueness statement needs a precise ambient measure. The family space $\mathcal{V}$ lies in a proper affine subspace of the full matrix-family space because of the column-sum constraints. Consequently, it has zero Lebesgue measure in the unrestricted ambient space. The claim that non-unique projections occur on a Lebesgue-null set should therefore specify Lebesgue measure relative to the affine hull of $\mathcal{V}$, or an equivalent intrinsic measure. Otherwise the statement is formally vacuous.

The Moreau-decomposition discussion adds little in its present form. The manuscript correctly avoids applying Moreau's theorem directly to a potentially nonconvex contingent cone. Once it replaces that cone with the closed convex Clarke tangent cone, however, the projection residual is already known to lie in its polar, so the decomposition is essentially the trivial decomposition into zero tangential component plus the residual. This passage could be shortened unless it is used to derive a further model-specific result.

In the proof of Proposition 4.4, the statement that M'_k is column-stochastic should explicitly use $1^\top R = 1^\top$, with dimensions indicated. The present calculation suppresses this first step.

Lemma 4.8 is correctly stated, but "weight independence of the strata" may be broader than what is shown. The zero sets $\mathcal{F}_d$ and the resulting rank are weight-independent; the actual Whitney stratification selected for $\mathcal{F}_d$ is not discussed. "Weight independence of the zero sets and rank" would be more precise.

Example 5.2 should keep the superscript "inhom" prominent. In the homogeneous category, the dimension-matched model set is generally a strict subset of the flat families.

The notation $D = \sum_k N_k$ depends on the chosen realizing process, not solely on V. Different joint processes can have the same two-time conditional family but different sets of positive-probability histories. The theorem should either say "for any chosen realizing process" or state the universal bound as the quantity determined solely by n and m.

"Generic" is carefully defined as almost everywhere rather than open-dense or Zariski-generic, which is welcome. That care should be retained when the following sentence changes to genericity in the ED-degree sense, since the two uses of "generic" refer to different frameworks.

Proposition 7.1 is sound, but its interpretation should retain the hidden-clock qualification.
The inclusion

$$\mathcal{F}_d^{\mathrm{hom}} \subseteq \mathcal{F}_d^{\mathrm{inhom}}$$

and the resulting profile inequality are immediate and correct. The strict dimension-matched example is also valid. However, the broader claim that time-inhomogeneity substitutes for hidden dimension should be read alongside Theorem 5.7: the homogeneous class can itself simulate time dependence by encoding the time layer within the hidden state. The comparison is therefore between explicit time dependence and additional clock-carrying hidden states, not necessarily between genuinely stationary and nonstationary mechanisms.

The numerical section cannot presently support independent verification.
The paper refers to a self-contained script "available from the author," but does not provide a repository, persistent archive, supplementary file, or the explicit numerical families. The master seed alone is insufficient because reproducibility can depend on:

- the random-number generator and software version;
- the precise method used to sample uniformly from the simplex;
- the optimization parameterization and bounds;
- stopping criteria and tolerances;
- the deterministic child-seed construction;
- the exact LP input and rational certificate. The script, numerical inputs, environment information, and rational certificates should be deposited with the paper.

"Certified upper bounds" is potentially too strong for the optimizer outputs. Evaluating the objective at an exactly feasible lift does provide an upper bound on h_d. In the reported calculations, however, feasibility and stochasticity are represented in floating-point softmax coordinates. Stationarity of L-BFGS-B does not certify global optimality, and floating-point evaluation does not by itself constitute a rigorous numerical certificate. "Computed feasible upper bounds" or "numerical upper bounds" would be safer unless interval arithmetic or exact feasibility verification is supplied.

The heading "Recovery of hidden dimension" overstates the result. The experiments establish exact vanishing at a known generating dimension. They do not establish that this is the smallest possible dimension because no certified positive lower bound is obtained below it. The text acknowledges this correctly, but the subsection heading and some statements in the abstract imply actual rank recovery. A more accurate heading would be "Vanishing at the generating dimension" or "Numerical evidence for hidden dimension."

The positive values below the generating dimension do not certify minimality. The author correctly notes that the entries 0.062434, 0.010136, and 0.089160 are optimizer upper bounds. A failed

search for a smaller value is evidence, but it cannot exclude a zero or substantially smaller minimum. The phrase "the evidence is unambiguous" is therefore too strong. The large scale separation between these values and the directly evaluated zeros is encouraging, but numerical separation is not a proof of infeasibility.

An optimizer candidate should not be called "the projection" unless global optimality is established. The manuscript repeatedly refers to "the projection at $d = 3$," "the $d = 2$ projections," and "the $d = 3, 4, 5$ projections." Multistart local optimization yields candidate nearest points, not certified metric projections. These should be called "best fitted points found," "reported optimizer candidates," or similar unless a global certificate is provided.

Positive block masses do not establish that the candidate lies in the interior of $\mathcal{F}_d$. The paper identifies some fitted points as interior because their minimal observed block masses are positive. Positivity places the lift parameters inside Θ_d^+, but it does not show that the compressed point lies in the relative interior of $\mathcal{F}_d$. The compression map may have deficient rank, the point may belong to a lower-dimensional image stratum, or the same observable family may have multiple representations. Interior or stratum membership requires analysis of the differential and local image, not merely positive masses.

Chart stationarity does not automatically verify the stratum orthogonality condition. A small gradient in the softmax parameter chart establishes stationarity with respect to the parameter directions explored by that chart. It is equivalent to orthogonality to the full tangent space of the relevant model stratum only if the compression map is locally regular and its differential spans that tangent space. The paper does not establish these conditions. The claim that equation (9) is numerically verified should therefore be weakened or supported by a rank calculation for the compression Jacobian.

Extremely small fitted block masses are not proof of intrinsic boundary contact. Values such as 10^{-23}, 10^{-54}, and 10^{-27} show that the optimizer drives a particular parameterization toward degeneracy. They do not prove that the limiting observable point lies on the boundary of $\mathcal{F}_d$, because the same compressed family may admit another nondegenerate representation. They also do not prove that the optimizer has found the global projection. The numerical result should be described as strong parameter-level boundary-seeking behaviour rather than certified boundary contact.

The marginal-consistency example is valuable, but the exact certificate should be supplied. The computation of $\kappa_c(V) = 0.276613$ is the strongest numerical result because it gives a dimension-independent lower bound through Theorem 5.6. However, the claimed exact rational verification cannot be assessed from the paper. The exact matrices, primal solution, dual solution, and equality of the certified objectives should be included in supplementary material.

The phrase "realizability plateau" should be used cautiously.

The theorem certifies only

$$h_d(V) \geq 0.195595$$

uniformly in d. The reported values decrease substantially from 1.807963 to 1.153167 over $d = 2, \ldots, 5$, so the computations do not yet exhibit a numerical plateau in the ordinary sense. What is proved is a strictly positive dimension-independent floor. "Certified realizability gap" would be more precise.

The gauge computation illustrates coordinate dependence but does not itself "confirm" Corollary 3.4. The two different constrained minima obtained from representatives of the same orbit show that the determinant-constrained surrogate is not orbit-invariant. Corollary 3.4 is already an exact theoretical result about continuous orbit-invariant functionals. The experiment illustrates the failure of this particular surrogate rather than confirming the corollary itself.

The determinant-floor experiment needs fuller specification. The paper does not state clearly whether the constraint is imposed on every g_i, how the admissibility inequalities are enforced, what parameterization is used for mass-preserving invertible matrices, or how the optimizer handles nearly singular and very large entries. These details are especially important because the reported condition numbers and entries exceed 10^3.

The selection-bridge open problem is not well defined when projections are non-unique. Open Problem (i) refers to "the compression of its projection onto $\mathcal{F}_d$." A nearest point need not be unique. The

conjecture must either quantify over all projections, select one by an additional rule, or be stated only where the metric projection is unique. Moreover, equality on the zero stratum is largely automatic because $V \in \mathcal{F}_d$ projects to itself.

Open Problem (ii) incorrectly treats h_n as an established contraction of Δ_V. Example 5.2 identifies h_n with distance to the flat set, while Proposition 5.3 compares this distance with a norm of Δ_V through a Łojasiewicz inequality. It does not construct h_n as a contraction or natural functional of Δ_V. Different families can have the same deficit data while having different distances to the flat set, unless an additional result is proved. The proposed classification should therefore not cite h_n as an example of a deficit contraction without defining and establishing that relationship.

The lumping operation in Open Problem (v) has not been defined. For conditional kernels, coarse-graining is generally law-dependent and cannot be represented by a simple matrix product unless the appropriate conditional distributions are specified. The notation ΛV should be defined, including the compatibility conditions and the role of the process law, before posing a decomposition of $h_n(\Lambda V)$.

The connection between compactification and finite termination should not be overstated.

Constructing a compactified lift model whose image equals each fixed $\mathcal{F}_d$ would restore exact-realization language at that dimension. It would not automatically imply that

$$\bigcup_{d \geq n} \mathcal{F}_d$$

is closed or that every asymptotically liftable family belongs to some finite-dimensional model. These are related questions, but the latter requires a uniform control on dimension that fixed-d compactification alone does not provide.

The references do not yet adequately position the paper within the modern latent-variable literature.

The paper cites two positive-realization sources and the classical Kemeny-Snell text, but the proposed model sets also connect directly with:

- algebraic statistics of hidden Markov models;
- identifiability and minimal-state realization;
- probabilistic automata and finitary processes;
- nonnegative rank and tensor-factorization methods;
- model-order selection for latent-state processes;
- compatibility and marginal problems for collections of conditional distributions. A broader literature discussion is necessary to substantiate the novelty of "memory rank" and distance to $\mathcal{F}_d$.

The phrase "one representative labeling per ordered block-size class" should be explained and justified. The reader needs to know which ordered compositions of d were included and why hidden-state permutations make all remaining labelings equivalent.

The number of free parameters, optimizer bounds, convergence thresholds, and treatment of softmax redundancies should be reported. Softmax charts contain additive non-identifiabilities that can affect Hessians and stationarity diagnostics.

"The optimizer located the generating lift unprompted" is useful empirical information but depends on what counts as locating it. The corresponding parameter distance or observable residual criterion should be specified.

The statement that the transformed family "visibly approaches" reset form is not supported by a figure. Either provide a plot or replace "visibly" with a numerical norm estimate.

The data-and-code statement should give a permanent link, archive identifier, software environment, and licence rather than saying only that the script is available from the author.

The manuscript states that a self-contained verification script capable of regenerating all reported quantities is available from the author. It would strengthen the reproducibility of the numerical results if this script, together with the computational environment or dependency information and the exact LP certificates, were deposited in a persistent public repository or supplied as supplementary material.